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CODESLAYER 2K26

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Project Title : AI-Powered Construction Material Accountability System

Team Name : HACKNOVA

Team Members : Yash Gupta , Vishnu Kumawat , Tarun Singh , Shubham Makkar

Tracks Chosen: *(AI/ML , Internet of things , Open Innovation)* :

PROBLEM STATEMENT & TARGET AUDIENCE

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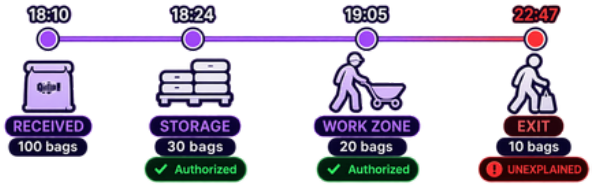
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4-30%
Potential material waste/loss*

₹ 4200
Potential exposure from one unexplained movement
ILLUSTRATIVE DEMO SCENARIO
10 cement bags = 4200

24/7
CCTV generates footage
But human review doesn't scale

- **Construction sites know what materials were purchased and recorded, but cannot continuously verify what physically happens to them.**
- **CCTV captures movement while inventory tracks transactions, creating a visibility gap. This causes unexplained material loss, financial exposure, budget overruns, project delays, and accountability issues.**
- **BuildGuard AI connects physical movement with digital records.**



Imagine a project manager watching costs rise while materials mysteriously disappear. BuildGuard AI helps managers, engineers, contractors, and inventory teams track every movement, detect anomalies, prevent losses, and maintain accountability across construction sites.

OUR UNIQUE SOLUTION

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BUILDGUARD AI CONNECTS CCTV FOOTAGE WITH INVENTORY RECORDS TO TRACK MATERIAL MOVEMENT, IDENTIFY UNEXPLAINED ACTIVITY, AND ASSESS FINANCIAL RISK.

IT DETECTS WHAT MOVED, WHERE, WHEN, AND WHETHER IT WAS AUTHORIZED. THE SYSTEM HIGHLIGHTS INVENTORY MISMATCHES, CALCULATES RISK SCORES, GENERATES ALERTS WITH EVIDENCE, AND QUANTIFIES POTENTIAL LOSSES FOR BETTER ACCOUNTABILITY.

MATERIAL INVENTORY
Real-time view of materials tracked across the site.

MOVEMENT / ALERTS
AI detects unaccounted movement and flags potential issues.

FINANCIAL IMPACT
Estimates potential financial exposure. Requires verification.

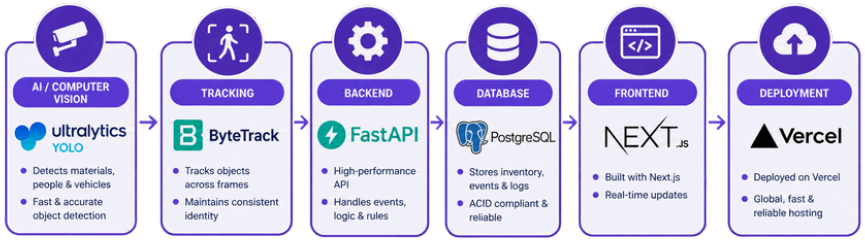
ACCOUNTABILITY
Identifies missing records and tracks accountability status.

An AI-powered system that transforms construction-site CCTV into actionable material intelligence.

TECH STACK + ARCHITECTURE

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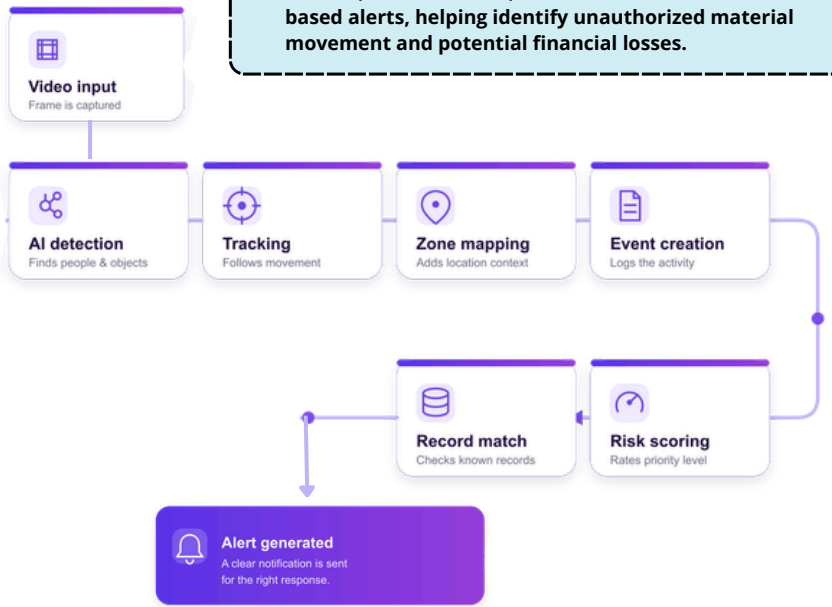
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BuildGuard AI's complete technology architecture, from CCTV/video capture to AI detection, object tracking, zone-event analysis, inventory reconciliation, risk scoring, and alerts.

Our tech stack combines YOLOv8 for AI-based object detection, ByteTrack for real-time tracking, FastAPI for backend processing, PostgreSQL for reliable data storage, Next.js for the dashboard, and Vercel for deployment. Together, these technologies enable scalable, secure, real-time monitoring, inventory reconciliation, risk analysis, and evidence-based alerts.

It tracks materials, people, vehicles, zones, quantity, and movement. Events are matched against inventory records, scored for risk, and converted into evidence-based alerts, helping identify unauthorized material movement and potential financial losses.



FEASIBILITY AND SHOWSTOPPERS

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BuildGuard AI is feasible using existing CCTV infrastructure, AI vision, object tracking, cloud computing, and inventory systems. YOLOv8 and ByteTrack enable material detection and tracking, while FastAPI and PostgreSQL support scalable processing. The solution requires minimal additional hardware and can progressively integrate with construction sites and existing digital workflows.

EXPECTED IMPACT

15–25% Less material loss and unexplained variance	**70%+** Less manual review and reconciliation time
Real-time Material visibility movement and exposure, live	**100%** Explainable alerts with evidence and impact

THE CHALLENGE Material movement is hard to verify in real time

4–30% Material loss / waste can affect construction projects	**₹45,200** Potential exposure per material-loss incident*
24 / 7 Sites never stop but CCTV review is still manual	**Visibility gap** Physical vs. digital records are not reliably connected

Key challenges include poor camera angles, low lighting, occlusion, incomplete CCTV coverage, and inaccurate inventory records. AI detection or tracking errors can create false alerts, while real-time processing across multiple cameras may require significant computing resources. Data privacy, system integration, network reliability, and user adoption also need careful consideration.

USP(UNIQUE SELLING PRICE) & BUSINESS MODEL

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Busuness Model

BuildGuard AI follows a B2B SaaS subscription model, charging construction companies based on sites, cameras, and monitoring needs. Additional revenue comes from premium analytics, cloud storage, integrations, and enterprise support. Free pilots can attract customers, while recurring subscriptions and scalable cloud infrastructure ensure sustainable revenue and long-term market growth.

Unique Selling Point

“We connect what the camera sees with what the inventory expects.”
BuildGuard AI bridges the gap between physical material movement and digital inventory records, automatically detecting unexplained movements, reconciling them with expected transactions, and generating evidence-based risk alerts.

REFERENCES

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- Research Papers & Academic Journals
- Official Documentation & Technical Resources
- Open-Source Libraries & GitHub Repositories
- APIs & Developer Documentation
- Datasets & Data Sources
- Industry Reports & Case Studies
- Other Relevant Online Resources

Important Pointers

(ToBeDeletedBeforeSubmission)

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Your Team Name

1. You may change the current layout (boxes/sections), but ensure all required content is included.
2. Keep your submission to **a maximum of 6 slides**, including the **Title Slide**.
3. It’s recommended to put your ideas in **bullet points**.
4. Focus on the **core idea** – keep it crisp and avoid over-explaining.
5. **Do not change the title bar** (project name, team, etc.).
6. Ideas must be **original and not copied** from existing solutions.
7. Save and submit your file as a **PDF** only. Other formats will not be accepted.